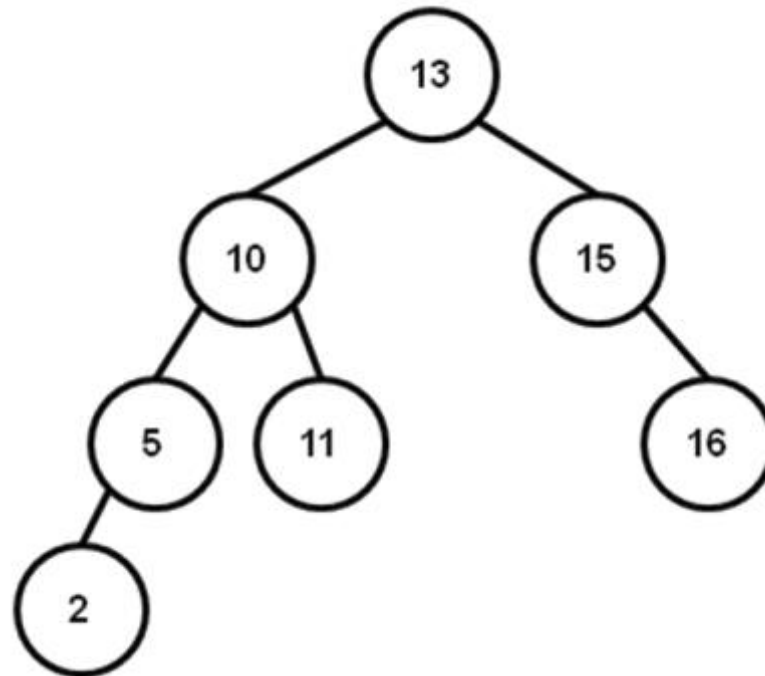
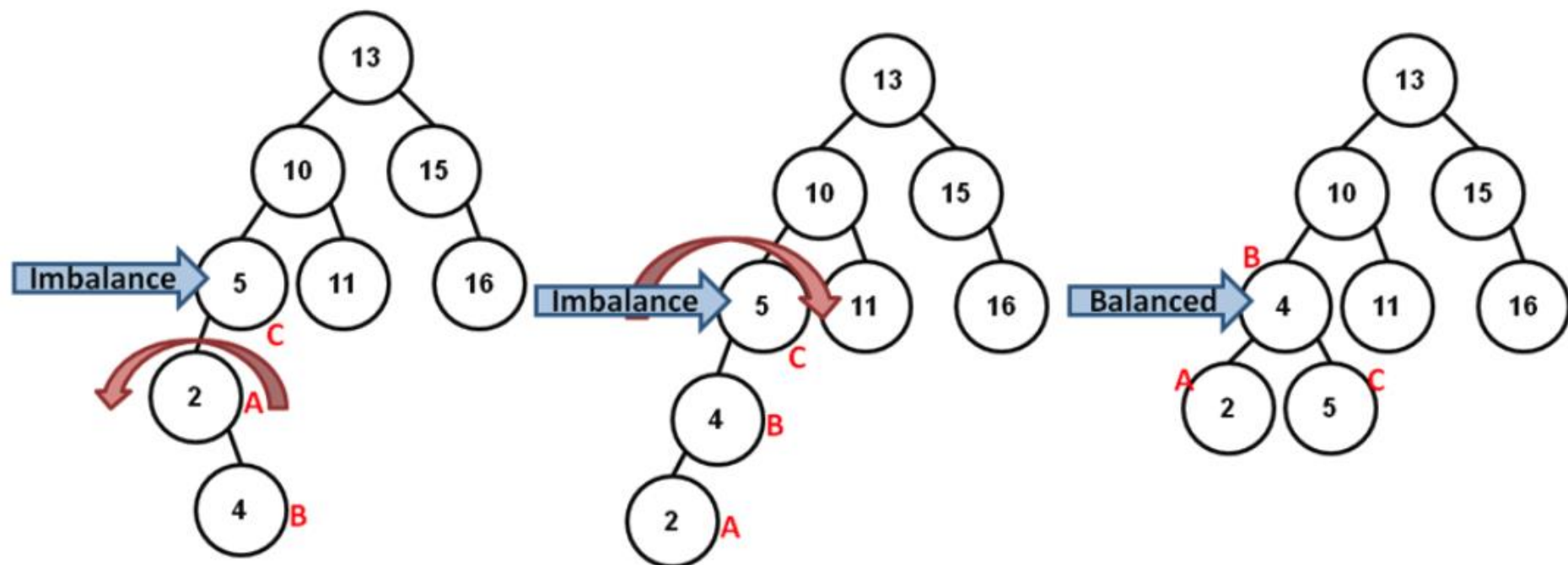


1. Insert 4 into the AVL Tree

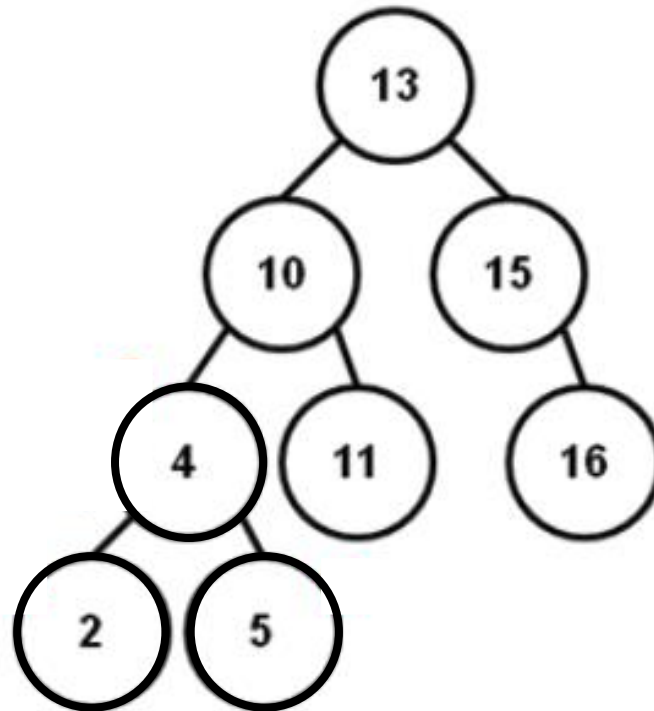
- Insert 4 into the AVL tree below. If the tree becomes unbalanced, balance the tree and redraw the final tree after the insertion.

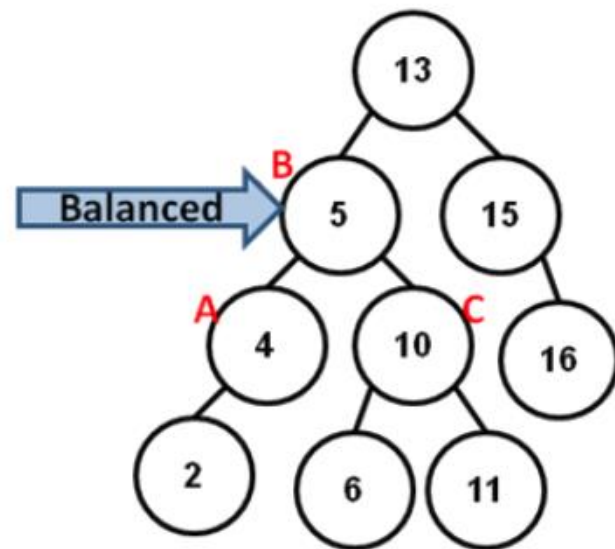
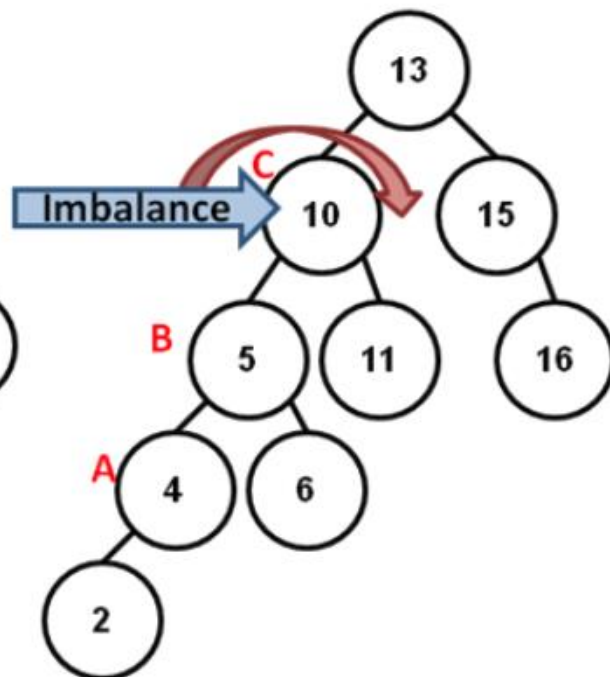
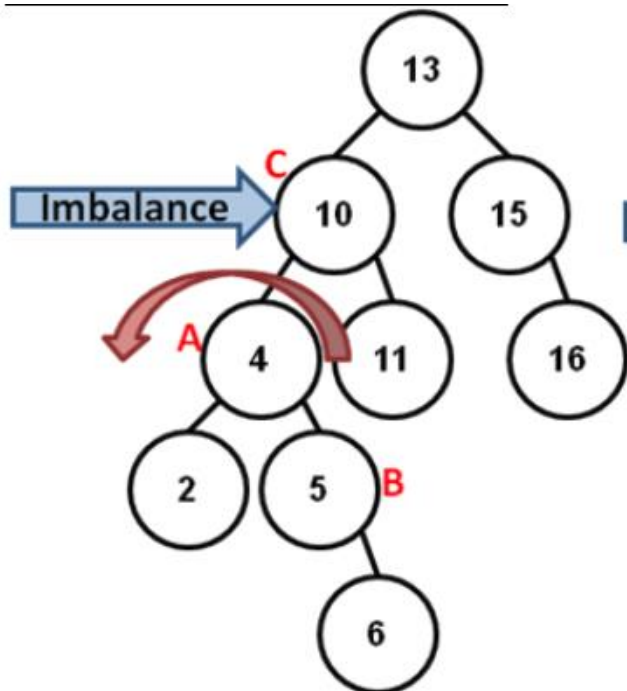




2. Insert 6 into the AVL Tree

- Insert 6 into the AVL tree below. If the tree becomes unbalanced, balance the tree and redraw the final tree after the insertion.





3. Construct an AVL Tree

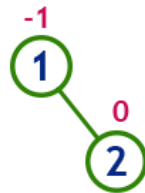
- Construct an AVL Tree by inserting numbers 1, 2, 3, 4, 5, 6, 7, 8.

insert 1



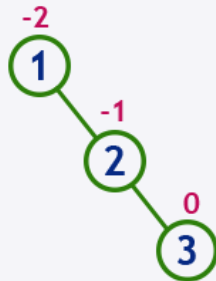
Tree is balanced

insert 2

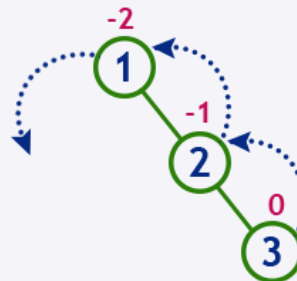


Tree is balanced

insert 3

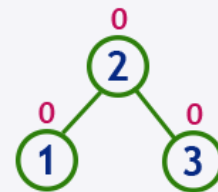


Tree is imbalanced



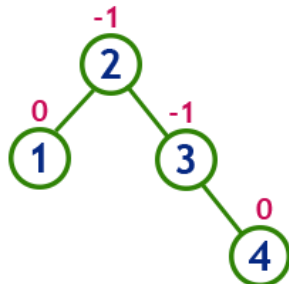
LL Rotation

After LL Rotation



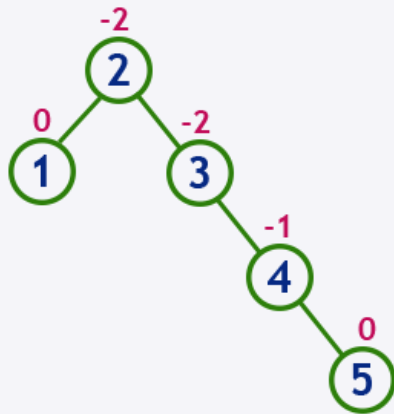
Tree is balanced

insert 4

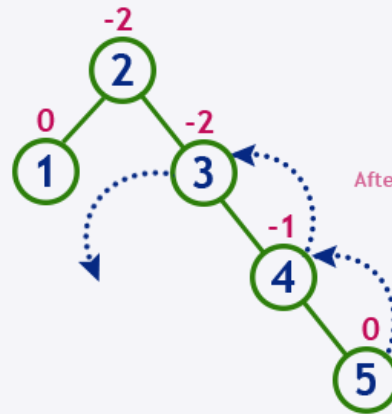


Tree is balanced

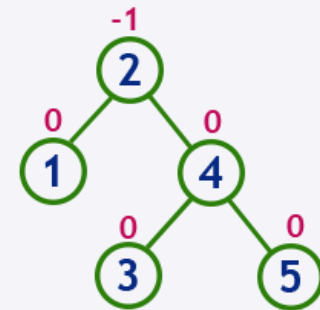
insert 5



Tree is imbalanced

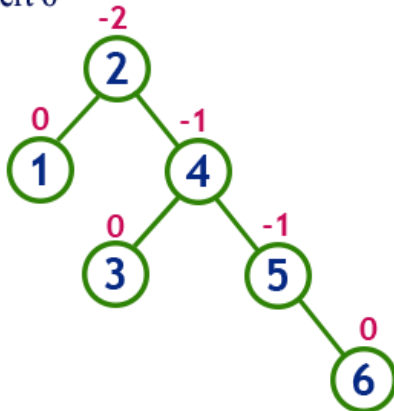


After LL Rotation at 3

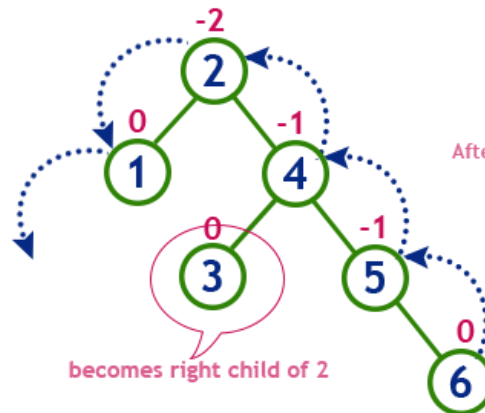


Tree is balanced

insert 6

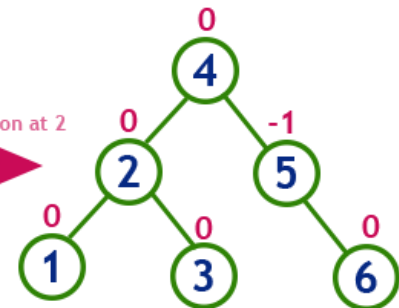


Tree is imbalanced



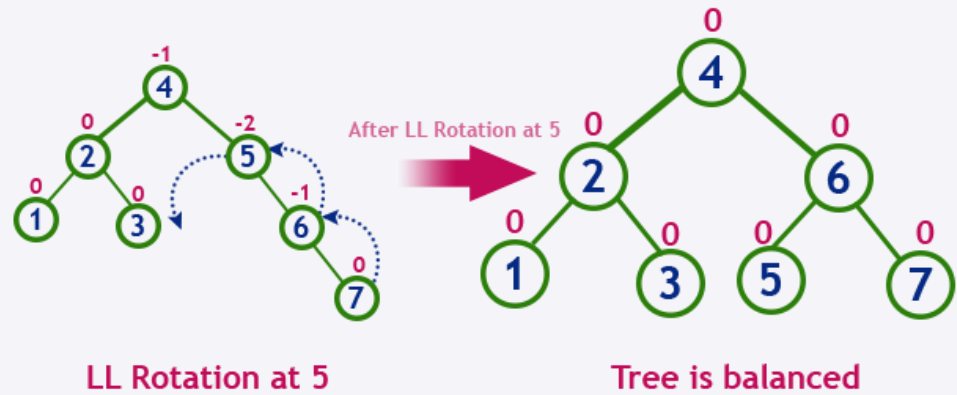
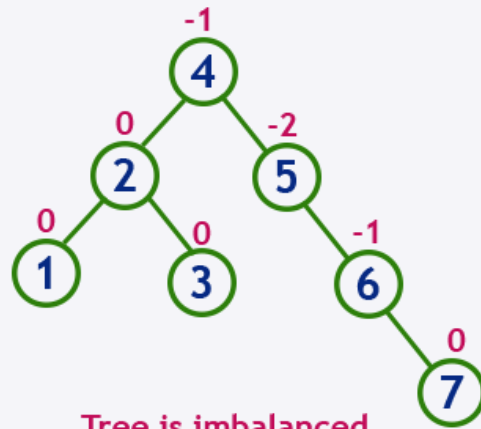
becomes right child of 2

After LL Rotation at 2



Tree is balanced

insert 7



insert 8

